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- c. within **a the plane of symmetry of RV-HANGER situated between the two largest faces of RV-HANGER**, connect two non-intersecting bodywork sections.

C3.5 Floor Bodywork

C3.5.1 Floor Body

“Floor Body” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BODY.
- b. when viewed from below, fully obscure:
 - i. RV-PU-ICE
 - ii. RV-DIFF.
- c. have up to two sections in any Z-Plane.

C3.5.2 Floor Foot

“Floor Foot” Bodywork must:

- a. lie in its entirety within RV-FLOOR-FOOT.
- b. when viewed from above, fully obscure RS-FLOOR-FOOT.
- c. have up to two sections in any Z-Plane.

C3.5.3 Floor Sidewall

“Floor Sidewall” Bodywork must:

- a. lie in its entirety within RV-FLOOR-SIDEWALL.
- b. have up to two sections in any Y-Plane.
- c. **forward of $X_R = 250$ and** above $Z = 100$, have up to two sections in any Z-Plane.

C3.5.4 Main Floor

“Main Floor” results from the Trim and Combination of the following components:

- a. Floor Body.
- b. Floor Foot.
- c. Floor Sidewall.

A Fillet Radius, no greater than 30 mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Main Floor must:

- d. be a single, simply connected volume, with no overlapping regions.
- e. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.
- f. when viewed from the side, fully obscure RS-FLOOR-SIDEWALL.

Main Floor Aerodynamic Surfaces must:

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- g. contain no radius of curvature less than 25mm, except for regions:
 - i. within 5mm of the plan-view-boundary of Main Floor, when viewed from above or below.
 - ii. within 45mm of a single point that lies in RV-FLOOR-SPHERE.
 - iii. of convex curvature, visible from above, and within 10mm of the load application points defined in Article C3.18.6.
 - iv. within 5mm of the Fillet Radii permitted as part of the Main Floor Trim and Combination, where the radius of curvature must be no less than 5mm.

C3.5.5 Floor Board

“Floor Board” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BOARD.
- b. have up to three sections in any X, Y and Z-Plane.

Floor Board Aerodynamic Surfaces visible from the side must:

- c. rearward of $X_F = 825$, at any point, in any Z-Section have a:
 - i. normal with a positive X component.
 - ii. tangent that forms an angle of at least 15 degrees to the X-Axis.
- d. contain no concave radius of curvature less than 100mm.

Furthermore:

- e. compliance with (c) and (d) is not required within 35mm of up to three points, which must sit forward of $X_F = 1100$ and be no closer than 50mm to each other.

C3.5.6 Floor Bib

“Floor Bib” Bodywork must:

- a. lie in its entirety within RV-FLOOR-BIB.
- b. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.

Floor Bib Aerodynamic Surfaces must:

- c. contain no concave radius of curvature:
 - i. visible from below.
 - ii. less than 50mm, if visible from above.

C3.5.7 Floor Leading Edge Device

“Floor Leading Edge Device” Bodywork must:

- a. lie in its entirety within RV-FLOOR-LED.
- b. have up to ~~five~~ **three** sections in any X and Z-Plane.

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- c. above $Z = 60$, be no less than 4mm thick when measured normal to Floor Leading Edge Device Aerodynamic Surfaces only.

Floor Leading Edge Device Aerodynamic Surfaces above $Z = 60$ must:

- d. contain no convex radius of curvature less than 2mm

C3.5.8 Floor Winglet

“Floor Winglet” Bodywork, which is exempt from Article C3.2.4, must be fitted and results from the Trim and Combination of the following components:

- a. RS-FLOOR-WINGLET.
- b. **“Winglet Profile”**, which must:
 - i. lie in its entirety within RV-FLOOR-WINGLET.
 - ii. contain no more than a single section in any X, Y and Z-Plane.
- c. A single Hanger, which must:
 - i. when viewed from below, be fully obscured by RS-FLOOR-WINGLET.

Furthermore, Winglet Profile Aerodynamic Surfaces must:

- d. contain no concave radius of curvature less than 20mm.

Once Trimmed and Combined, Floor Winglet and the annotated surface ‘*TRIM-SURFACE*’ may only be modified together, and in the following order:

- e. Rotated by ± 5 Degrees about a Y-aligned axis that passes through the point $[X_R, Z] = [295, 200]$
- f. Translated by ± 10 mm in Z.

Furthermore, Floor Winglet may be:

- g. trimmed below the annotated surface ‘*TRIM-SURFACE*’ in RS-FLOOR-WINGLET.

Once fully defined, Floor Winglet must:

- h. have up to two sections in any Z-Plane below the annotated surface ‘*TRIM-SURFACE*’ in RS-FLOOR-WINGLET.
- i. ~~except for C3.5.8(c)~~, lie in its entirety within RV-FLOOR-WINGLET-ENVELOPE.

C3.5.9 Floor Fence

“Floor Fence” Bodywork must:

- a. lie in its entirety within:
 - i. an axis aligned cuboid of the following dimensions $[X, Y, Z] = [450, 40, 200]$
 - ii. RV-FLOOR-FENCE.
- b. be fully visible from either above or below, such that any surface obscured from one direction is visible from the other.
- c. have up to two sections in any Y-Plane.

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Furthermore,

- d. when considering a section formed by the intersection with any Z-Plane, within 100mm of the rearmost point of the section, a line tangent to any part of the section visible from the side must not form an angle greater than 5 degrees, when measured against the X-axis.

C3.5.10 Floor Corner

“Floor Corner” Bodywork is exempt from Article C3.2.4, but must:

- a. lie in its entirety within RV-FLOOR-CORNER.
- b. have up to 3 sections in any X-Plane.
- c. have up to 4 sections in any Y-Plane.
- d. have up to 2 sections in any Z-Plane.

C3.5.11 Floor Tyre IR Sensor

“Floor Tyre IR Sensor” Bodywork must:

- a. lie in its entirety within a freely positioned instance of RV-FLOOR-TYRE-IR.
- b. not lie within RV-FLOOR-CORNER.
- c. be fully obscured by RS-FLOOR-BODY when viewed from above.
- d. fully enclose an infra-red sensor for measuring tyre temperature.

Floor Tyre IR Sensor Aerodynamic Surfaces must:

- e. contain no radius of curvature less than 20mm.

C3.5.12 Floor Bodywork Assembly

“Floor Bodywork Assembly” results from the Trim and Combination of the following components:

- a. Main Floor.
- b. Floor Bib.
- c. Floor Leading Edge Device.
- d. Floor Fence.
- e. Floor Board.
- f. Floor Winglet.
- g. Floor Corner.
- h. Floor Tyre IR Sensor, if fitted.

Before trimming, the following must be discarded:

- i. any Floor Fence remaining above Main Floor.
- j. any Floor Board remaining below Main Floor.
- k. any Floor Winglet remaining inboard of Floor Sidewall.
- l. any Main Floor remaining within RV-FLOOR-CORNER.

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m. any Floor Tyre IR Sensor remaining below Main Floor.

A Fillet Radius, no greater than 25mm, may be applied along the intersections between the remaining component parts.

Once fully defined, Floor Bodywork Assembly must:

- n. be a single volume, with no overlapping regions.
- o. when viewed from below, fully obscure RS-FLOOR-BODY.
- p. when viewed from below, be coincident with the surfaces of RV-FLOOR-BODY that are visible from below and directly beneath:
 - i. RS-FLOOR-REF.
 - ii. RS-FLOOR-STEP.
- q. forward of $X_F = 1500$, when viewed from the rear, fully obscure Floor Leading Edge Device.
- r. when viewed from the side, fully obscure RS-FLOOR-BOARD.

C3.5.13 Floor Bodywork Group

Once the components defined in Articles C3.5.1 to C3.5.12 have been constructed in accordance with these provisions, including any sub-assembly operations, the resulting union is defined as “**Floor Bodywork**”.

C3.5.14 Floor Auxiliary Components

Once Floor Bodywork has been fully defined and Final Assembly in Article C3.12 completed, the following “**Floor Auxiliary Components**”, which are exempt from Article C3.2.4 (b), may be fitted:

- a. Up to three “**Floor Body Stays**”, which may be designed to only take load in tension, and each of which must:
 - i. have a single inboard attachment ~~location point anywhere on the Sprung Mass except Floor Bodywork~~ that lies between $X_F = 1775$ 2500 and $X_R = 275$ 250, inboard of $Y = 400$ and above $Z = 175$.
 - ~~ii. be fixed on its inboard end to the entirely sprung part of the car.~~
 - ii. ~~be fixed on its outboard end to~~ have a single outboard attachment point that lies on Floor Bodywork between $X_F = 2500$ and $X_R = 250$.
 - iii. have a circular cross section and lie in its entirety within a 5mm diameter right circular cylinder, except within 25mm of the inboard and outboard attachment points, or within 10mm of any adjustment mechanism.
 - iv. when viewed from below, be fully obscured with Floor Bodywork in place.
 - v. not intersect any surfaces of Bodywork, except within 25mm of the inboard and outboard attachment points, and within 7mm of up to two points which lie on the Engine Cover.
- b. Up to eight Hangers, each of which must:
 - i. intersect Floor Board.

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- ii. lie no closer than 50mm to any other Hanger.
 - iii. when viewed from the side and below, be fully obscured with Floor Bodywork in place.
 - iv. lie in its entirety within the union of RV-FLOOR-BOARD and RV-FLOOR-FOOT.
- c. A single “**Floor Board Brace**”, which must:
 - i. lie in its entirety within a 40mm diameter right circular cylinder and RV-FLOOR-BRACE, except within 2mm of Forward Chassis and 35mm of the intersection point of the right circular cylinder’s axis and Forward Chassis.
 - ii. be rigidly fixed at its inboard end to Forward Chassis.
 - iii. be rigidly fixed at its outboard end to Floor Board.
 - iv. when viewed from the side, be fully obscured with Floor Bodywork in place, except within 2mm of Forward Chassis and 35mm of the intersection point of the right circular cylinder’s axis and Forward Chassis.
- d. A single fairing, if necessary, to shroud the single device permitted in Article C12.7, which must:
 - i. be fully within RV-BIB-STAY and be symmetrically arranged about $Y = 0$.
 - ii. in any Z-plane contain only one section.
- e. A single “**Floor Foot Stay**”, which may be designed to only take load in tension, must:
 - i. have a single inboard attachment point that lies no more than 25mm distant from the intersection of Floor Board Brace and Forward Chassis, or in the absence of a Floor Board Brace, within RV-FLOOR-BRACE.
 - ii. be fixed at its inboard end to Forward Chassis.
 - iii. be fixed at its outboard end to Floor Foot.
 - iv. when viewed from the side, be fully obscured with Floor Bodywork in place.
 - v. lie in its entirety within a 10mm diameter right circular cylinder, except within 25mm of the inboard and outboard attachment points, or within 10mm of any adjustment mechanism.

Furthermore, once all Floor Auxiliary Components have been defined:

- f. they must be Trimmed and Combined with any Bodywork they intersect.
- g. a Fillet Radius no greater than 20mm, may be applied along the intersections between Floor Auxiliary Components and any other Bodywork 5mm may be applied along the intersections between Floor Body Stays and any other Bodywork. Fillet radii, not greater than 20mm, may be applied along the intersections between any other Floor Auxiliary Component and any other Bodywork.

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C3.6 Plank and Skids

C3.6.1 Plank Assembly

A “**Plank Assembly**” must be fitted, consisting of the Plank, Skids and mountings.

The following provisions apply to the Plank Assembly:

- a. The upper surface of the Plank Assembly must lie at $Z = 0$, and must not be in contact with the External Air Stream.
- b. The geometry of the Plank Assembly must conform to RV-PLANK with a general manufacturing tolerance of $\pm 0.5\text{mm}$ when new, unless a specific dimensional tolerance is defined.
- c. The Plank Assembly must have three 34mm diameter holes whose axes are parallel to the Z-axis. The centres must lie on $Y = 0$ and have X-locations:
 - i. $X_F = 500\ 800$,
 - ii. $-600 \geq X_C \geq -800$, and on or behind $X_F = 1000$.
 - iii. $470 \leq X_{PU} \leq 630$. Furthermore, the centre of the rearmost hole must lie and on or ahead of $X_R = -500$.
- d. The thickness of the Plank Assembly measured normal to the lower surface must be $10\text{mm} \pm 0.2\text{mm}$ and must be uniform when new. A minimum thickness of 8mm will be accepted due to wear, and conformity to this provision will be checked at the peripheries of the holes required by (c) above.

Evaluation of Skid wear will be made by physical measurement of the Skid with it removed from the car. The Measurement will be made with a micrometer with a 6.35mm diameter anvil. When measured in this way, the entire periphery of each hole must be within the specified tolerance.

Four additional 10mm diameter holes are permitted provided their sole purpose is to allow access to the bolts which secure the FIA ADR to the Survival Cell.

C3.6.2 Plank

The following provisions apply to the “**Plank**”:

- a. The material of the Plank is defined in C15.3.5. ~~free, but it must be homogeneous with a specific gravity between 1.3 and 1.45, or if pocketed be made from a bonded assembly the upper 0.5mm of which must have a specific gravity of between 1.3 and 1.65 and the remainder, excluding pockets, be made from a homogeneous material with a specific gravity of between 1.3 and 1.45.~~
- b. The Plank may comprise no more than three pieces, the forward one of which may not be any less than ~~900mm~~ 600mm in length.
- c. Between $X_F = 630\ 930$ and $X_C = -800$ and behind $X_C = -400$, pocketing of the Plank is permitted between $Z = -7$ and $Z = -0.5$. The periphery of any pocket, in any Z-Plane, must be not less than 10mm from either the edges of the Plank or any holes or recesses in the Plank. In any vertical plane, the internal pocket Fillet Radii must be at least 3mm and in any Z-Plane, must be at least 10mm. Pockets may only be filled with a material having a specific gravity of less than 0.25.

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C3.6.3 Skids

The lower surface of the Plank may be fitted with flush mounted Skid material which can only be fitted in place of plank material. “Skids” may only be fitted around each of the 3 holes defined in C3.6.2.c and must conform to the geometries specified below.

A metallic Skid is mandatory around the forward-most Plank hole. A metallic Skid is optional around the middle and rearward-most Plank holes. If a metallic Skid is not present, it may be either; replaced with a separate piece of Plank material, or integrated into the Plank. The resulting geometry must contain the hole and counterbore detail present in RV-SKID-MID and RV-SKID-REAR.

- a. The Skid material around the forward-most hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-FWD and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
 - iii. fastened directly to the Front Floor Structure with zero degrees of freedom
- b. If metallic, the Skid material around the central hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-MID and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
- c. If metallic, the Skid material around the rear-most hole in the plank must:
 - i. be made to the geometry defined by RV-SKID-REAR and associated drawing.
 - ii. be divided into three separate parts as defined in the Reference Volume.
- d. Metallic Skids must be made from materials as defined in C15.3.6 either:
 - ~~iii. Titanium alloy Ti6Al4V (according to AMS4928 or AMS4911) in annealed condition. No additional heat treatment is permitted; or~~
 - ~~iv. 17-4PH stainless steel (to AMS 5604 / ASTM A 693 or AMS 5643/ ASTM A 564). When the skid is new, its hardness must be no more than 38 HRC.~~

~~Furthermore, they may only be machined from solid and no processes (such as forging, rolling, welding, or coating) may be carried out either before or after machining.~~

The material to be used in each Competition is defined in FIA-F1-DOC-041.

The only permitted deviation from the specified geometry is the removal of minimum material for the fasteners.

C3.6.4 Plank and Skid Mountings

The Plank, and Skids must be fixed to the car using fasteners which are no smaller than M6 and are made from grade 12.9 or 10.9 steel.

If used to attach a Skid to the car, the number of fixings for each type of Skid is defined as follows:

- a. RV-SKID-FWD: three fixings in each of the two forwardmost resulting Skids and two fixings for the rearward most resulting Skid.